

5(a)	from graph $\ln Q = 2.9$ (so $Q = 18.2 \mu\text{C}$)	B1
	$C = Q / V$	C1
	$= 18.2 / 12 = 1.5 \mu\text{F}$	A1
5(b)	gradient $= -0.25$	C1
	gradient $= -1 / RC$	C1
	$R = 1 / (0.25 \times 1.5 \times 10^{-6})$ $= 2.7 \times 10^6 \Omega$	A1
	or $\frac{Q}{Q_0} = e^{-t/CR}$ or $\ln Q - \ln Q_0 = \frac{-t}{CR}$	(C1)
	e.g. $\frac{4.95}{18.2} = e^{-5.2 / (1.5 \times 10^{-6} R)}$ or $1.6 - 2.9 = 5.2 / (1.5 \times 10^{-6} R)$	(C1)
	$R = 2.7 \times 10^6 \Omega$	(A1)

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Question	Answer	Marks
5(c)	$W = \frac{1}{2} QV$	C1
	$= \frac{1}{2} \times 18.2 \times 10^{-6} \times 12$	A1
	$= 1.1 \times 10^{-4} \text{ J}$	
	or $W = \frac{1}{2} CV^2$	(C1)
	$= \frac{1}{2} \times 1.5 \times 10^{-6} \times 12^2$	(A1)
	$= 1.1 \times 10^{-4} \text{ J}$	
5(d)	or $W = \frac{1}{2} Q^2 / C$	(C1)
	$= \frac{1}{2} \times (18.2 \times 10^{-6})^2 / 1.5 \times 10^{-6}$	(A1)
	$= 1.1 \times 10^{-4} \text{ J}$	
5(d)	straight line with different negative gradient starting from (0, 2.9)	M1
	straight line between $t = 0$ and at least $t = 5.0\text{s}$ with twice the gradient of the original line	A1